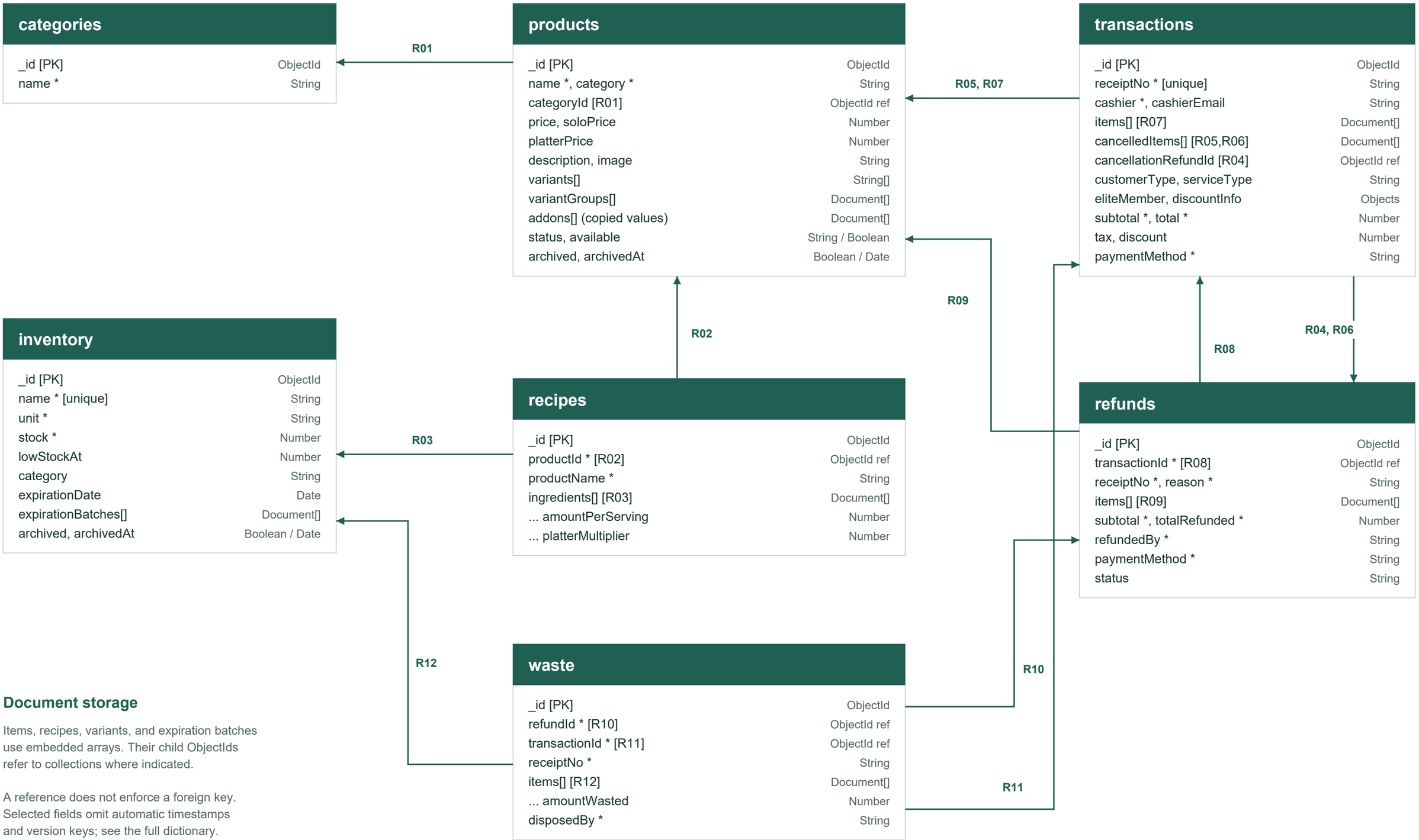
Figure 10A. Sales and ingredient records

Collection overview. Selected fields are shown; the data dictionary contains all 318 field paths, including embedded records.



Document storage

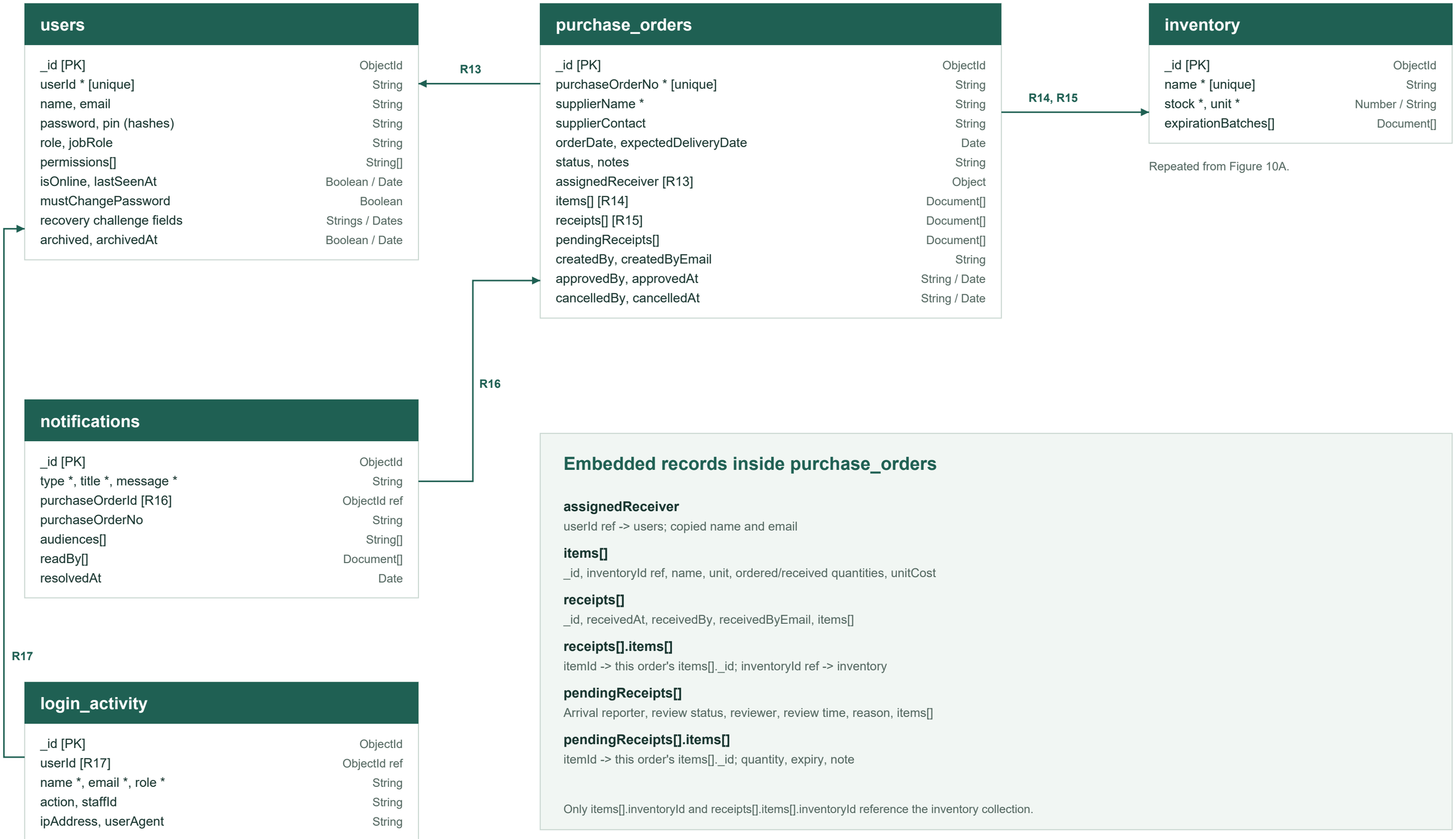
Items, recipes, variants, and expiration batches use embedded arrays. Their child ObjectIds refer to collections where indicated.

A reference does not enforce a foreign key. Selected fields omit automatic timestamps and version keys; see the full dictionary.

Arrows: stored reference -> target collection. Rxx: relationship register on page 4. *: schema-required field.

Figure 10B. Accounts and procurement

Receiver assignments, purchase-order lines, delivery receipts, notifications, and account activity.



Embedded records inside purchase_orders

assignedReceiver

userId ref -> users; copied name and email

items[]

_id, inventoryId ref, name, unit, ordered/received quantities, unitCost

receipts[]

_id, receivedAt, receivedBy, receivedByEmail, items[]

receipts[].items[]

itemId -> this order's items[]._id; inventoryId ref -> inventory

pendingReceipts[]

Arrival reporter, review status, reviewer, review time, reason, items[]

pendingReceipts[].items[]

itemId -> this order's items[]._id; quantity, expiry, note

Only items[].inventoryId and receipts[].items[].inventoryId reference the inventory collection.

Arrows: stored reference -> target collection. Rxx: relationship register on page 4. *: schema-required field.

Figure 10C. Catalog and supporting records

These five collections declare no ObjectId references to other models. Text snapshots do not create reference relationships.

addons

_id [PK]	ObjectId
name *	String
nameKey * [unique]	String
price	Number
active, archivedAt	Boolean / Date

Product add-ons copy names and prices.

discounts

_id [PK]	ObjectId
name *	String
nameKey * [unique]	String
percentage * (0-100)	Number
scope (item / order)	String
active, archivedAt	Boolean / Date

Transactions store discountInfo snapshots.

email_verifications

_id [PK]	ObjectId
email * [unique]	String
otpHash *	String
expiresAt * [TTL]	Date
attempts	Number

TTL index uses expireAfterSeconds = 0.

Records are eligible for expiry at expiresAt.

forecastsnapshots

_id [PK]	ObjectId
dataset * [compound unique]	String
targetDate * [compound unique]	String date
source * (demo / real)	String
predictedSales *	Number
generatedAt *	Date
trainedThrough *	String date
method	String

Unique key: dataset + targetDate. Automatic timestamps are disabled.

system_audit

_id [PK]	ObjectId
module *, action *	String
entityId, entityName	String snapshots
actor, actorEmail	String snapshots
details	String
changes	Mixed
createdAt, updatedAt	Date

entityId is a string with a module context; no Mongoose ref is declared.

Storage boundaries for the paper

Elite Member details are embedded in transactions; saved member templates live in browser localStorage.

Supplier names and contacts are embedded in purchase orders; supplier entry history also lives in browser localStorage.

There is no separate Member, Supplier, Payment, OrderItem, DeliveryReceipt, or StockMovement model in this implementation.

Stock movement history is represented through audit events; sales and refund quantities also reside in their respective records.

All 16 collections have generated document identifiers and version keys. Category and ForecastSnapshot have no automatic timestamps.

Declared ObjectId relationship register

The 17 references below correspond to the arrows in Figures 10A and 10B. MongoDB does not enforce their target existence.

ID	SOURCE COLLECTION / FIELD	TARGET COLLECTION	REQUIRED	STORAGE
R01	products.categoryId	categories._id	No	Single reference
R02	recipes.productId	products._id	Yes	Single reference
R03	recipes.ingredients[].inventoryId	inventory._id	No	Embedded array
R04	transactions.cancellationRefundId	refunds._id	No	Single reference
R05	transactions.cancelledItems[].productId	products._id	No	Embedded array
R06	transactions.cancelledItems[].refundId	refunds._id	No	Embedded array
R07	transactions.items[].productId	products._id	No	Embedded array
R08	refunds.transactionId	transactions._id	Yes	Single reference
R09	refunds.items[].productId	products._id	No	Embedded array
R10	waste.refundId	refunds._id	Yes	Single reference
R11	waste.transactionId	transactions._id	Yes	Single reference
R12	waste.items[].inventoryId	inventory._id	No	Embedded array
R13	purchase_orders.assignedReceiver.userId	users._id	No	Single reference
R14	purchase_orders.items[].inventoryId	inventory._id	No	Embedded array
R15	purchase_orders.receipts[].items[].inventoryId	inventory._id	Yes	Embedded array
R16	notifications.purchaseOrderId	purchase_orders._id	No	Single reference
R17	login_activity.userId	users._id	No	Single reference

Relationship interpretation

A required reference field must contain an identifier when its containing document exists; target existence is an application concern.

One target may be referenced by many source documents. Array references may point to multiple targets from one source document.

No unique index is declared on recipes.productId; a one-recipe-per-product database guarantee must not be claimed.

Receipt itemId values point to items[]._id inside the same purchase order; they are not links to a separate collection.

Only selected attributes appear in the figures. Data_Dictionary.md and database_schema.json retain the full 318-path structure.